

Quad Watch Winder

K Karel

VIEW IN BROWSER

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Summary

This watch winder is inspired by this thing: https://www.thingiverse.com/thing:2021221 Thank you Simone Dassi!

Hobby & Makers > Electronics

Tags: arduino watch winder atmega328p watchwinder

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Thank you Simone Dassi!
I adopted the shape of watch holder and plug. I modelled all parts in Fusion 360 from scratch.

ATMEGA is programmed like "Arduino on breadboard" as seen here: https://www.arduino.cc/en/Tutorial/ArduinoToBreadboard - the minimal version without external clock, running on 8 MHz.

I put everything on a piece of prototype PCB. I desoldered all the pins from motor drivers and soldered leads directly on them. Do not forget to close the jumper pads on the driver.
I also removed all the resistors from driver pcbs - to shut off the LEDs. Could have removed the LEDs but resistors seemed to be easier option.

All pcbs are mounted in the winder with double sided mounting tape. USB connector is glued in with superglue and everything is then covered with

midplate. USB PCB needs to be trimmed in order to fit in. I filed it down from both sides carefully. Base is glued together with superglue.
All other mounts are done with M3 screws and M3 square nuts.

Every watch holder can be set to run counter clockwise, clockwise or be off. It runs 10 minutes and then waits 50 minutes.

Setting the winder:
After programming the chip, nothing turns. Press button for more than 0.5 s. Press again with short press. The LED flashes - top left winder is set to run CCW. Press again with short press. The LED flashes twice - top left winder is set to run CW. Press again - no LED. You can scroll the CCW-CW-idle as many times as you want. Then long press sets the desired program for the winder. Another series of short presses scroll the settings for top right winder. Long press to set it. Another series of short presses for bottom left, confirmed with long press and the same for bottom right one. It looks complicated but it is not.
Long - short - short - short ... set with long - short - short .... set with long - short - short .... set with long - short ... set with long. Winder starts moving after the last long press.
It can be set as many times as you want. Settings are stored in EEPROM.

Parts Used:

Motors and drivers: http://s.aliexpress.com/JVjyIzaQ

ATMEGA328P-PU: http://s.aliexpress.com/JrMNBRjM

16mm LED Light Push Button: http://s.aliexpress.com/uMJjymma
Pick the 5V version. Do not forget to buy the socket for pushbutton.

Power adapter with MicroUSB socket: http://s.aliexpress.com/32ueyamm

Double Sided Prototype PCB Breadboard: http://s.aliexpress.com/7nEbeIZv

MICRO USB To DIP Adapter: http://s.aliexpress.com/VFbmaeAv

28 pin IC Socket
10 M polarised capacitor
0.1 M ceramic capacitor

M3x10 screws with half round head 20 pcs
M3x10 countersunk screws 4 pcs
M3 square nut 16 pcs
Bearing 623 inner diameter 3mm, outer diameter 10 mm, width 4 mm 8 pcs

Print instructions Category: DIY Print Settings

Printer Brand: Prusa

Printer: I3 MK3S

Rafts: No

Supports: No

Resolution: 0.3 mm

Infill: 15%

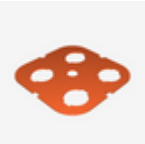
Filament: Fillamentum PLA VertigoGrey Post-Printing

Holes for mounting the bearings should be tapped with M3 tap.

Model files



base_1.stl



midplate.stl



box.stl



holder.stl



plug.stl



base_2.stl



base_3.stl



cover.stl

Find source .stl files on Thingiverse.com

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